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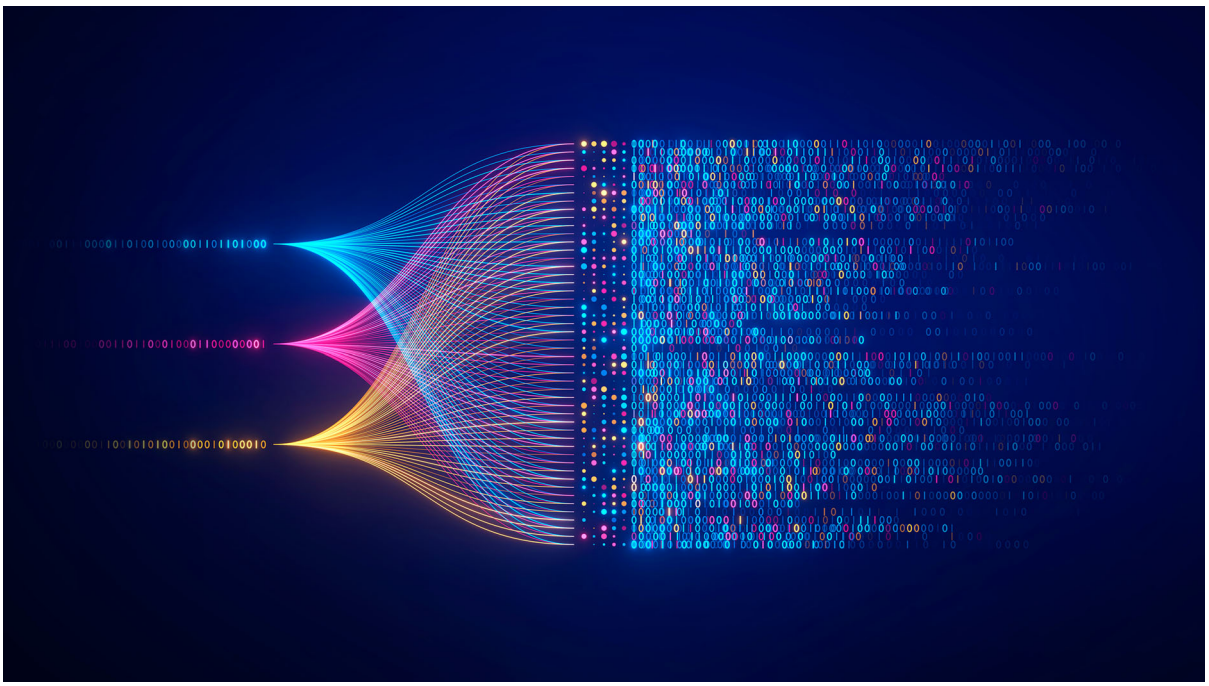

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Attention-based reading, highlighting, and forecasting of the limit order book

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Managing high-frequency data in a limit order book (LOB) is complex due to its high dimensionality, irregular timing, and complex spatiotemporal dependencies across price levels. These challenges often exceed the capabilities of conventional time-series models. Accurate prediction of the multi-level LOB, not just the mid-price, is crucial for understanding market dynamics but is difficult due to the interdependencies among attributes like order types, features, and levels. This study introduces advanced sequence-to-sequence models to forecast the entire multi-level LOB, including prices and volumes. Our key contribution is a compound multivariate embedding method that captures spatiotemporal relationships. Empirical results show that our method outperforms others, achieving the lowest forecasting error while maintaining LOB structure.

Keywords: Limit order book; Multivariate time-series forecasting; Attention mechanism; Spatiotemporal embedding

JEL Classifications: G10, G17

1. Introduction

A limit order book (LOB) is a digital record of all outstanding buy and sell orders for a particular financial instrument, such as a stock or currency. It contains information about the price and quantity of each order, as well as the time at which the order was placed. The LOB continuously updates as new orders are submitted, executed, or canceled, providing an up-to-the-moment snapshot of market supply and demand. This evolution of orders highlights the distribution of orders across various price levels, providing information on market depth and potential price movements.

However, predictive modeling of LOBs presents considerable challenges due to the high frequency and complexity of the data. The dynamic nature of the LOB, with orders being added, modified, or removed at millisecond intervals,

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requires rapid processing and analysis of vast amounts of information. Additionally, LOB data has a complex multivariate time-series structure with compound attributes, where levels, types, and features are interrelated and jointly define each value. For instance, the ordinal structure in multi-level prices highlights the interconnectedness of these attributes. Therefore, accurately capturing the influence of each attribute is essential for effective predictive modeling of the LOB.

Another challenge in time series forecasting, including LOB prediction, is the non-stationarity of the data. Non-stationarity refers to changes in the statistical properties of a time series over time, such as mean, variance, or autocorrelation, which can significantly impact forecasting accuracy. Shifts in trends and structural changes over time can result in distributional differences between the training and test periods, undermining model performance. Techniques such as adaptive attention mechanisms (Liu *et al.* 2022) and trainable normalization methods (Kim *et al.* 2021) have been proposed to address these issues, but still struggle with abrupt changes in high-frequency data and require careful tuning to handle long-term dependencies.

Given the complexity and high dimensionality of LOB data, much of the existing literature has focused on forecasting the mid-price, which is defined as the average of the best bid and ask prices. The mid-price serves as a general indicator of market trends and is often used as a proxy for the market price at a given moment. A wide range of machine learning models have been applied to this task, including regression-based approaches (Ntakaris *et al.* 2018), support vector machines (Kercheval and Zhang 2015), and deep learning-based architectures such as convolutional and recurrent neural networks (Zhang *et al.* 2019, Tsantekidis *et al.* 2020). More recently, transformer-based models have also been explored for mid-price movement classification in LOB settings (Wallbridge 2020, Liu *et al.* 2024, Berti and Kasneci 2025), showing strong empirical performance in capturing local dependencies and temporal patterns in high-frequency market data.

Attention-based models like Transformer (Vaswani *et al.* 2017) and BERT (Devlin *et al.* 2019) excel at sequence-to-sequence prediction by assigning varying weights to different parts of the data, enabling them to focus on the most relevant information for more accurate predictions. These models have proven highly effective in tasks such as natural language processing (Cui *et al.* 2019, Galassi *et al.* 2020, Yang *et al.* 2021), speech recognition (Chorowski *et al.* 2015, Bahdanau *et al.* 2016, Zeyer *et al.* 2018), and time series forecasting (Fan *et al.* 2019, Shih *et al.* 2019, Du *et al.* 2020). Despite their success, applications of attention-based models in limit order book forecasting are still relatively limited. Recent studies, however, have started to explore this direction. For instance, Arroyo *et al.* (2024) used survival analysis and attention mechanisms to forecast order fill probabilities, Liu *et al.* (2024) applied vision transformers to classify mid-price movements, and Wu *et al.* (2024) employed transformer-based wavelet decomposition for short-term price movement prediction in the Chinese market. Additionally, Mäkinen *et al.* (2019) integrated jump diffusion models with attention-based neural networks to forecast jump arrivals. Nevertheless, none of these studies focused on predicting the full price and volume structure within the limit order book.

However, relying solely on mid-price movements is insufficient when a deeper understanding of market microstructure is required. As shown in figure 1, multiple LOB snapshots can share the same mid-price yet differ significantly in order depth and bid-ask spread (BAS). These differences highlight the limitations of mid-price-based approaches in contexts where liquidity, order imbalance, or spread variability play a central role. This concern is echoed by Sirignano (2019), which argues that modeling the joint distribution of bid and ask prices provides a more accurate and informative representation of LOB dynamics than assuming a fixed symmetric spread around the mid-price. While their approach incorporates bid and ask sizes as input features, it does not jointly predict these components. Fully capturing the structure of the LOB, including both price levels and their corresponding volumes, is essential for practical applications, such as in algorithmic trading and optimal execution strategies, where granular LOB information can significantly influence strategic decisions.

In this study, we present an attention-based approach utilizing the Spacetimeformer architecture (Grigsby *et al.* 2022) to address the unique challenges of LOB forecasting. Our method builds on previous research by introducing a customized embedding technique that effectively captures spatiotemporal features while maintaining the LOB's structural integrity, resulting in a more detailed and accurate depiction of market dynamics. By integrating both price and volume information, this approach significantly enhances predictive accuracy compared to traditional models, showcasing the strong potential of attention-based architectures to advance forecasting in high-frequency trading contexts. These advancements highlight the critical role of attention mechanisms in identifying complex patterns within LOB data, paving the way for more sophisticated and reliable market modeling.

Our contributions are three-fold: (1) we extend the capability of attention-based models beyond mid-price prediction, enabling them to capture and predict dynamic patterns of the LOB structure, including both prices and volumes; (2) we implement modified spatiotemporal embeddings that effectively integrate both spatial and temporal features, offering a model generalizable to multivariate time series with compound attributes; and (3) we improve forecasting accuracy by addressing non-stationarity in high-frequency financial data using a percent-change and min-max transformation method that does not add complexity to the model.

This paper is organized as follows: Section 2 offers an overview of the dataset and experimental setup. Section 3 details the methods used to address the challenges of LOB forecasting. Section 4 presents the experimental results, comparing our model's performance with various existing techniques. Section 5 discusses the implications of our approach, and Section 6 concludes the paper with suggestions for future research directions.

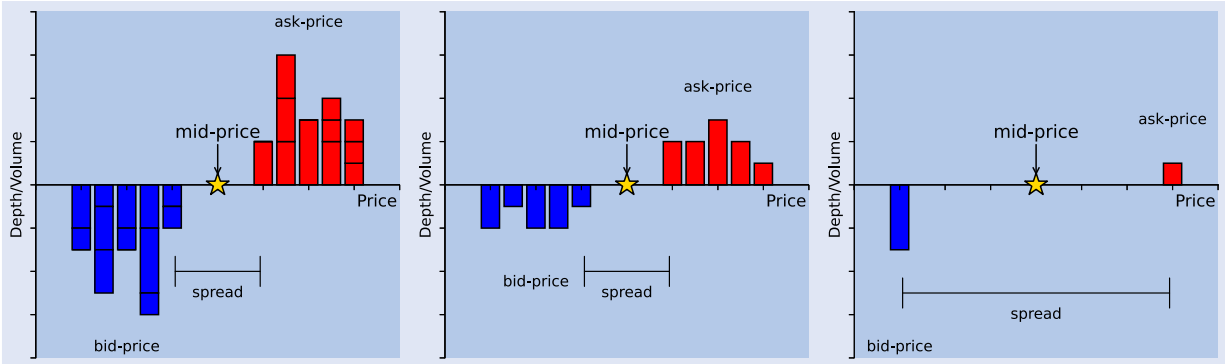

Figure 1. Three different LOB snapshots with the same mid-price: The three limit order book snapshots share the same mid-price, with the first and second snapshots showing varying levels of depth available, while the third snapshot features a larger bid-ask spread.

2. Data

The LOB is a real-time record of all buy and sell orders for a specific asset in financial markets. Organized by price levels, it displays the best bid (highest buy price) and best ask (lowest sell price) at any given moment, along with details like order volume and timestamps. In this section, we explore the high-frequency characteristics and spatiotemporal structure of the LOB data and outline our approach for formulating and transforming these inputs to support model development.

2.1. High-frequency dynamics and spatiotemporal correlations

2.1.1. High-frequency LOB. In high-frequency trading and market microstructure, traditional financial assumptions, such as stationarity and independent increments, are often considered inadequate by the real-time, high-frequency nature of the LOB. Unlike in conventional financial modeling, where these assumptions may hold over longer time horizons, the continuous flow of orders, frequent price changes, and rapid reactions by algorithmic trading result in time series data that are inherently noisy, irregular, and non-stationary. This volatility presents unique challenges in modeling and forecasting, as the traditional tools often used in lower-frequency financial data analysis do not sufficiently capture the nuanced behaviors inherent in high-frequency environments.

Figure 2 visually depicts these dynamics by showing rolling averages of AMZN stock's best-bid and best-ask prices over various time intervals–from seconds to hours–within a single trading day. While shorter time windows capture sharper price fluctuations and greater volatility, reflecting the underlying high-frequency dynamics, longer windows smooth out these short-term variations. This visual distinction underscores how time resolution significantly impacts our understanding of data. The choice of time scale in analyzing LOB data is crucial, as high-frequency behaviors may be masked in more aggregated views, potentially overlooking the critical, rapid adjustments inherent to LOB activity.

Further complicating the analysis of LOB data is the often-used assumption of a constant bid-ask spread. This simplification is commonly applied to streamline modeling efforts; however, figure 3 demonstrates that the bid-ask spread is far from constant in high-frequency trading. At smaller time intervals, spreads exhibit continuous and significant fluctuations, with notable changes occurring around critical times, such as market open and close. These time-dependent variations in both spread mean and volatility highlight the dynamic complexity of the LOB that a constant spread assumption cannot fully capture. While constant spread models offer mathematical tractability, they fail to represent the richness of real-time market dynamics, especially when considering the effects of microstructure in high-frequency settings (Abergel *et al.* 2016).

2.1.2. Spatiotemporal structure of LOB. The high-frequency nature of LOB reveals complex dynamics that influence the order book movements, with multiple components contributing to these fluctuations. In a multi-level LOB, each price level and corresponding volume plays an essential role as new orders are placed, canceled, or executed, reshaping the order book's structure and potentially impacting market behavior across different levels. These rapid and simultaneous changes reveal how the continuous interaction of market participants dynamically impacts both prices and volumes.

Figure 4 underscores this spatiotemporal relationship, presenting how mid-price movements (top panel) are shaped by changes in bid and ask prices and volumes of multiple levels. Price shifts in the multi-level structure reflect the cumulative effect of variations in both price and volume, which tend to be interconnected. In the middle and bottom heatmaps, we observe that large portions of the LOB remain static, showing 0% change, meaning that not all levels are altered with each order event. This sparse activity, where only specific levels undergo changes, suggests that most of the LOB remains stable between updates, with shifts happening intermittently across some levels. However, when a substantial movement occurs in a certain direction, it often influences neighboring levels, driving future movements in a similar direction. Notably, these movements often occur concurrently across various attributes, including price and volume, suggesting a compounded effect across the LOB. This sparse yet correlated activity provides valuable insight into how market depth and order flow dynamically shape the LOB across multiple levels.

These observations emphasize the need for a modeling approach that can accommodate the high-frequency,

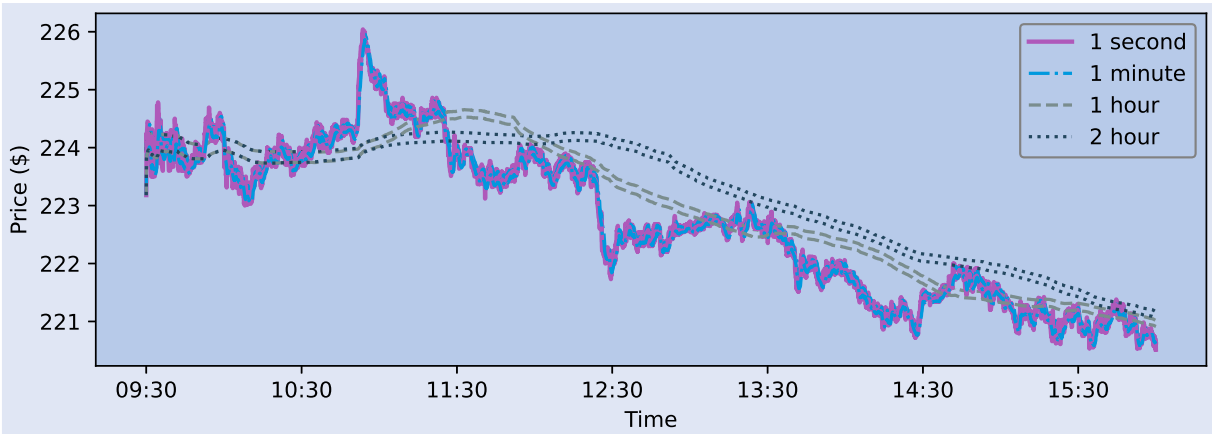

Figure 2. Rolling averages of AMZN stock's best-bid and best-ask prices across time windows ranging from 1 second to 2 hours. For each frequency pair, the upper line represents the best-ask price, while the lower line represents the best-bid price. Notably, sharp changes seen around 11:00 and 12:30 are smoothed in the 1- and 2-hour windows, where shorter-term fluctuations become less prominent.

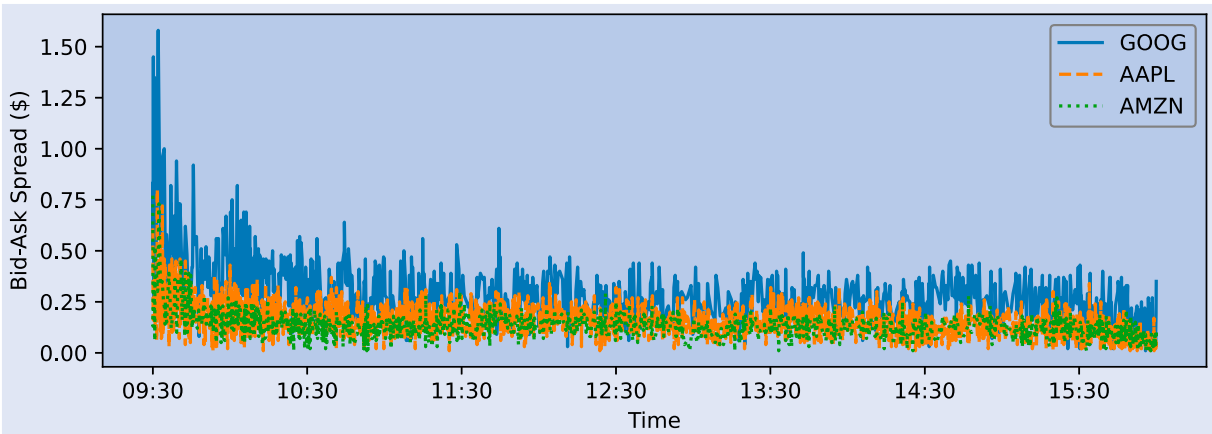

Figure 3. Bid-Ask Spread (BAS) for a selected trading day, calculated for three sample stocks: GOOG, AAPL, and AMZN.

time-varying, and spatiotemporal properties of LOB data. In this light, a data-driven method, rather than relying on traditional theoretical assumptions, is better suited to capture dependencies that evolve over short intervals. Adopting such an approach allows for a more accurate representation of the dynamics in the LOB, providing insights into market behavior that can inform effective decision-making in high-frequency trading contexts.

2.2. Data structure and preparation

As the first experiment, we utilize the LOBSTER dataset (Huang and Polak 2011), which offers real-world LOB data, including both message and order book information for selected stocks. We specifically focus on five major technology companies: AAPL, GOOG, INTC, MSFT, and AMZN. The sample covers the trading day of June 21, 2012, from the market opening at 9:30 am to the closing at 4:00 pm, capturing between 300 000 to 600 000 snapshots per stock, all recorded with millisecond time resolution at Level-5 of the order book. The order book file tracks the evolution of the LOB, while the message file logs the events that trigger updates, with timestamps ranging from milliseconds to nanoseconds.

For the second experiment, we evaluate the performance of the attention-based models on cryptocurrency LOB data. The dataset used for these experiments is sourced from Coinbase† and is downloaded from Kaggle.‡ It contains approximately 12 days of 1-minute interval LOB data for BTC, ETH, and ADA. A key difference from the stock LOBSTER dataset is that cryptocurrency markets operate continuously without trading halts, and the available data is aggregated at minute-level granularity, rather than at the event level as in the LOBSTER dataset. Summary statistics for both datasets, including frequency, price, and volume information, are presented in table 1.

The order book consists of two types of orders: bids (to buy) and asks (to sell). A LOB is continuously updated with every change in these orders, including new submissions, cancellations, deletions, or executions. Each snapshot of the LOB data is represented as $\vec{X}_i$ for $i \in \{1, \ldots, I\}$. Both bid and ask orders are captured as vectors containing the timestamp and price-volume pairs for each level $k \in \{1, \ldots, K\}$. Specifically, for the ith snapshot, $\vec{X}_i = [t_i, (p^b_{1i}, v^b_{1i}), \ldots, (p^b_{Ki}, v^b_{Ki}), (p^a_{1i}, v^a_{1i}), \ldots, (p^a_{Ki}, v^a_{Ki})]$, where t_i is the timestamp, $p^{b/a}_{ki}$ represents the level-k bid/ask price, and $v^{b/a}_{ki}$ represents the level-k bid/ask volume. Due to the price ordering in the multi-level structure of the LOB, the following

† coinbase.com
‡ https://www.kaggle.com/datasets/martinsn/high-frequency-crypto-limit-order-book-data

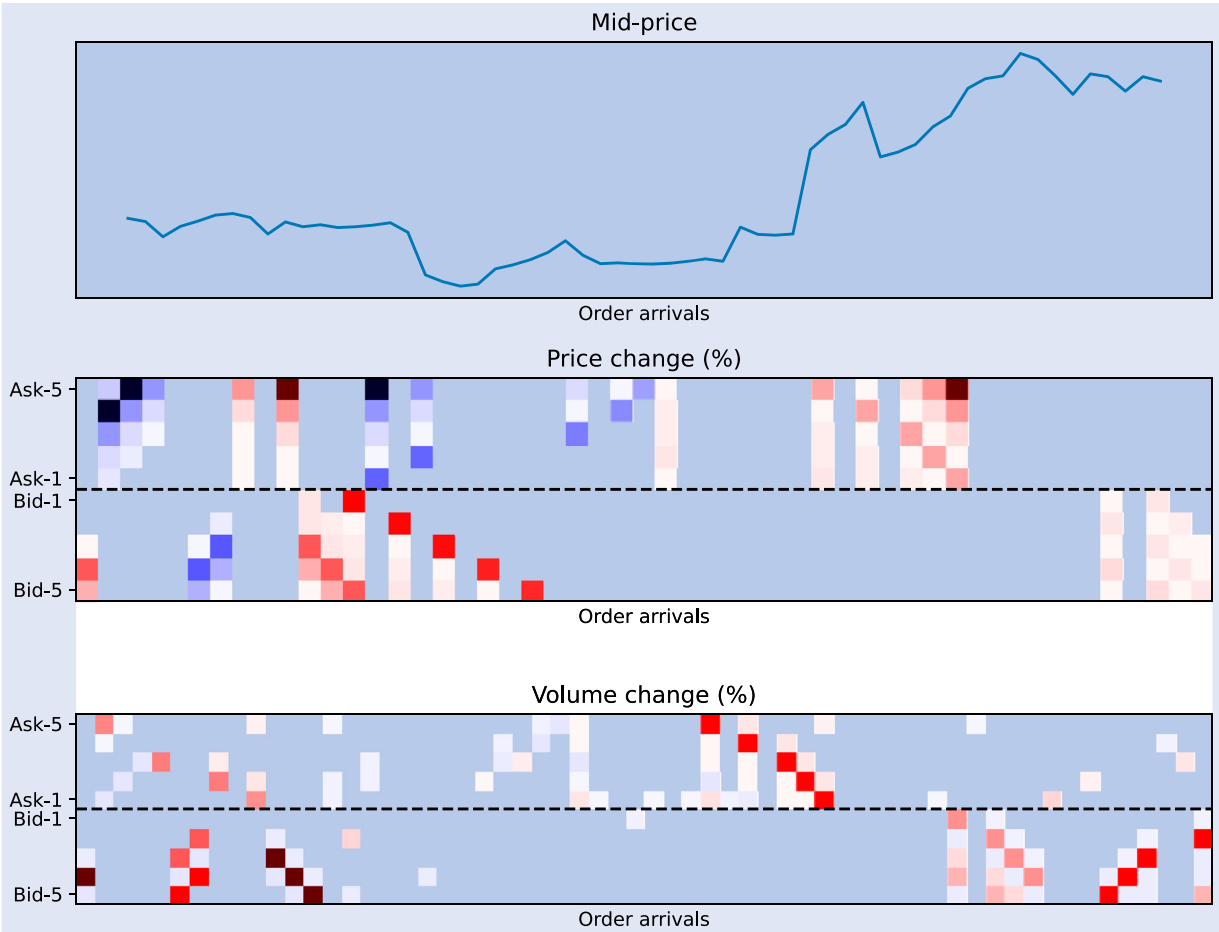

Figure 4. Mid-price (top), level-5 bid and ask price changes (middle), and level-5 bid and ask volume changes (bottom) aligned along the same order arrivals axis. In the middle and bottom heatmaps, darker shades indicate higher absolute percentage changes, with red representing upward movements and blue representing downward movements; White areas show no change. The mid-price is shown on a raw price scale, while percentage change values are scaled for better comparability across different scales. These plots illustrate data from the first 60 order arrivals of AAPL stock at market open.

Table 1. Summary statistics of stock and cryptocurrency LOB datasets across selected tickers.

			Mid-Price		Volume	
Ticker	Timestamps	Frequency	Low	High	Bid	Ask
Stock data						
AAPL	301 587	0.08	577.48	588.18	188	141
GOOG	112 673	0.21	563.73	579.82	161	129
INTC	581 030	0.04	26.62	27.59	16 550	14 020
MSFT	595 800	0.04	30.06	31.14	14 345	15 148
AMZN	155 935	0.15	220.52	226.03	249	139
Cryptocurrency data						
BTC	17 113	60.24	52 430.65	64 859.00	471 148	435 627
ETH	17 110	60.25	1937.43	2546.41	107 206	105 795
ADA	17 109	60.24	1.13	1.56	15 212	14 657

Notes: Frequency is the average time between snapshots (sec). Spread is the mean bid-ask gap (USD). Reported are the lowest and highest mid-prices, and average volumes at the best bid and ask levels. Prices are in USD; volumes in shares.

inequalities hold for each snapshot:

$$p^a_{k_1 i} < p^a_{k_2 i} \quad \text{for all } k_1 < k_2, \tag{1}$$

$$p^b_{k_1 i} < p^a_{k_2 i} \quad \text{for } k_1 = k_2 = 1, \tag{2}$$

$$p^b_{k_1 i} > p^b_{k_2 i} \quad \text{for all } k_1 < k_2 \tag{3}$$

The inequality (1) indicates that ask prices increase with depth, (2) ensures that the best bid is lower than the best ask, and (3) enforces that bid prices decrease with depth. To maintain consistency, we fix the number of levels, K, at 5 for both bid and ask orders in our experiments.

To ensure consistency and facilitate analysis, we select snapshots based on timestamps at equal 5-second intervals for stock data. For example, this results in approximately 4680 snapshots per trading day. We also concatenate data across all tickers, resulting in 100 dimensions for stock data and 60 for cryptocurrency data, where the total dimension is given by: number of assets * number of levels * 2 (bid/ask) * 2 (price/volume). The dataset is divided into training, validation, and testing sequences in a 6:2:2 ratio. For stock data sampled at 5-second intervals, a context window of 10 minutes is used to predict the next 2 minutes. For cryptocurrency data sampled at 1-minute intervals, a 2-hour context window is used to forecast the next 10 minutes.

Table 2. Performance metrics calculated for each input transformation across the top-5 price levels.

Input transformation	Percent-change	Min-max	Percent-change + Min-max
MSE	4.5447	0.0660	**0.0464**
MAE	1.3225	0.1469	**0.1260**

Notes: Mean Squared Error (MSE) and Mean Absolute Error (MAE) are provided for each method. All values have been descaled and are presented in their original units of dollars.

2.3. *Input transformation and scaling*

The input encoding pipeline for LOB prediction involves preprocessing the LOB data to be fed into a forecasting model (figure 5). This step involves extracting relevant features from the LOB data, such as the price, volume, and time of each order. The process begins by segmenting the Level-5 LOB data into sequences, each consisting of a 10-minute context window for input and a 2-minute target window for prediction. The pipeline then applies a set of embedding layers that encode different dimensions of the input. Specifically, time embeddings capture periodic and positional information, compound embeddings represent each input channel (e.g. price and volume at different levels), and context embeddings indicate whether a given timestamp belongs to the context window or the future target window to be predicted.

A key challenge in modeling LOB data is the presence of non-stationary features, which can hinder learning by causing distributional shifts between the training and testing periods. This is a common issue when working with long sequences of time series data. For instance, LOB prices often exhibit non-stationarity even within a single trading day, making it difficult for models to generalize across time.

In our experiment, we employ percent-change transformation of prices to facilitate training. The percent change transformation is applied to multivariate price sequences across all levels and for both bid and ask orders using the following equation:

$$p_{k,i}^{j\ \mathrm{perc}} = \frac{p_{k,i}^{j} - p_{k,i-1}^{j}}{p_{k,i-1}^{j}}, \tag{4}$$

where i denotes the snapshot index, j denotes bid or ask orders, and k denotes the price level. Figure 6 illustrates that the transformed series exhibits greater stationarity in price movements compared to the raw series. During training, we also apply min-max scaling to price and volume on a variable-wise basis to ensure that variables with large fluctuations do not disproportionately influence the model. The min-max scaling transformation for price and volume is defined as:

$$p_{k,i}^{j\ \mathrm{scaled}} = \frac{p_{k,i}^{j\ \mathrm{perc}} - \min_i(p_{k,i}^{j\ \mathrm{perc}})}{\max_i(p_{k,i}^{j\ \mathrm{perc}}) - \min_i(p_{k,i}^{j\ \mathrm{perc}})} \tag{5}$$

$$v_{k,i}^{j\ \mathrm{scaled}} = \frac{v_{k,i}^{j} - \min_i(v_{k,i}^{j})}{\max_i(v_{k,i}^{j}) - \min_i(v_{k,i}^{j})} \tag{6}$$

The performance of various transformation methods is compared in a later section, specifically in table 2 within Section 4.

3. Method

3.1. *Compound multivariate embedding*

The embedding process encodes time, values, variables, and context-target information. We employ the Time2Vec layer (Kazemi *et al.* 2019) to transform timestamps into frequency embeddings, which are then passed through a feed-forward layer to capture temporal patterns and data sequence. Additionally, a binary context-target embedding identifies whether a value is from the context or is to be forecasted, enabling the model to distinguish between observed and future data points.

Our data structure incorporates compound attributes such as multiple levels (level-k), types (bid or ask), features (price or volume), and stocks (tickers). To effectively capture dependencies among these attributes, we developed a compound multivariate embedding method inspired by the spacetimeformer approach (Grigsby *et al.* 2022). Unlike the original spacetimeformer, which assigns embedding integers to each variable separately, our method assigns embeddings to each attribute. Then, we combine and scale the multi-level embeddings. As shown in figure 7, each attribute within the variable dimension is encoded using a separate embedding layer. Specifically, one layer captures the order side (bid or ask), another encodes the feature type (e.g. price or volume), and a third represents the order level (e.g. level-1, level-2, etc.). When predicting multiple stock assets simultaneously, an additional embedding layer is included to differentiate between stock tickers (e.g. AAPL, AMZN, etc.).

By integrating and scaling these multiple embedding layers, our approach captures not only temporal and spatial patterns but also the complex relationships between different attributes within the LOB data. This method also reduces the number of parameters in the embedding layer while effectively incorporating interdependencies across attributes. The embedded input sequence is processed using an attention-based encoder-decoder architecture, as illustrated in figure 8. The encoder leverages multi-head self-attention modules to capture both local dependencies and global contexts across the entire sequence. This enables the model to learn complex interactions between price and volume movements across levels and time. The decoder is designed to generate future LOB states autoregressively. It incorporates masked self-attention, which ensures that predictions at each time step are based only on past and present information, preventing any information leakage from the future target window. Finally, a feed-forward layer is applied to project the decoder's output back into the original LOB format, yielding predictions for multi-level prices and volumes in the same structure as the input.

The attention mechanism (Vaswani *et al.* 2017) updates token representations and re-normalizes the resulting matrix:

$$\mathrm{Attention}(Q, K, V) \in \mathbb{R}^{L_x \times d} = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) V, \tag{7}$$

where query vectors $Q = W^Q X$, key vectors $K = W^K Z$, and value vectors $V = W^V Z$ are derived from learned parameters. For a sequence of length L and dimension d, the self-attention matrix outputs an $(L \times d)^2$ matrix, enabling assessment of both temporal and inter-variable dependencies.

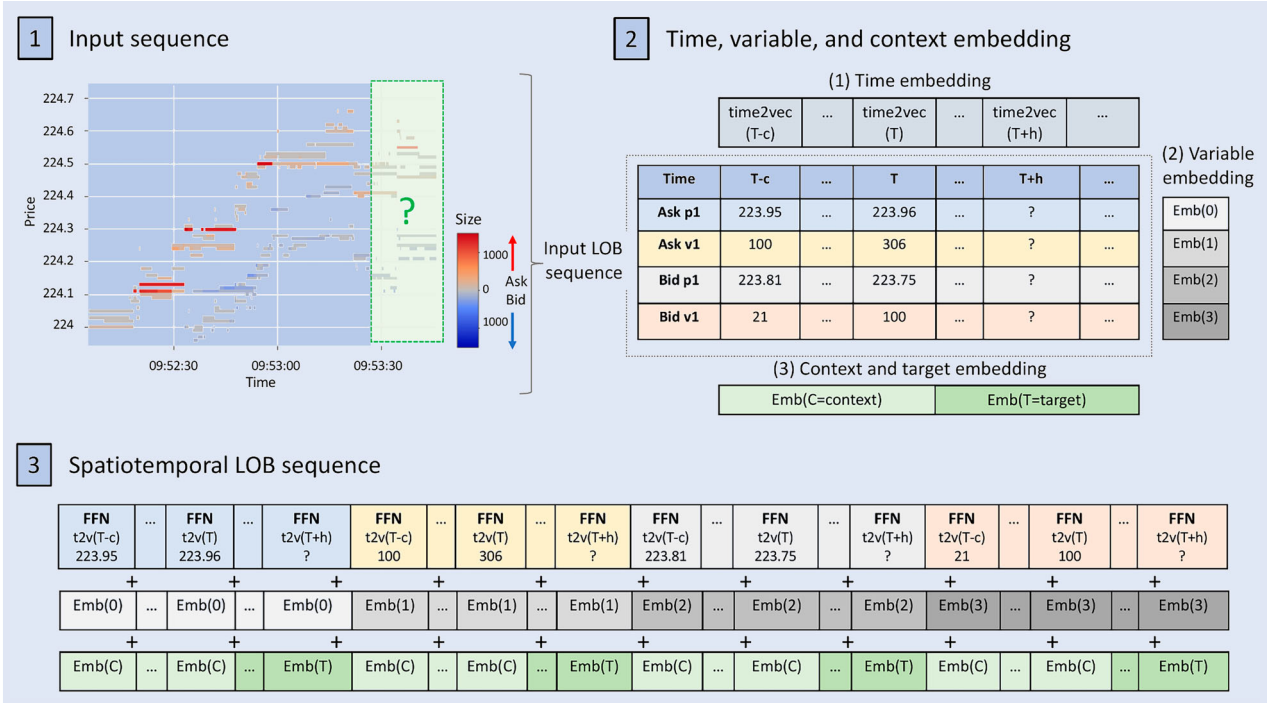

Figure 5. Input encoding pipeline for limit order book forecasting. (1) Level-5 LOB sequences are split into 10-minute context points and 2-minute target points. (2) Time, variable, and context embeddings are created to structure the input sequence. (3) The final LOB input sequence is generated by integrating these embedding layers.

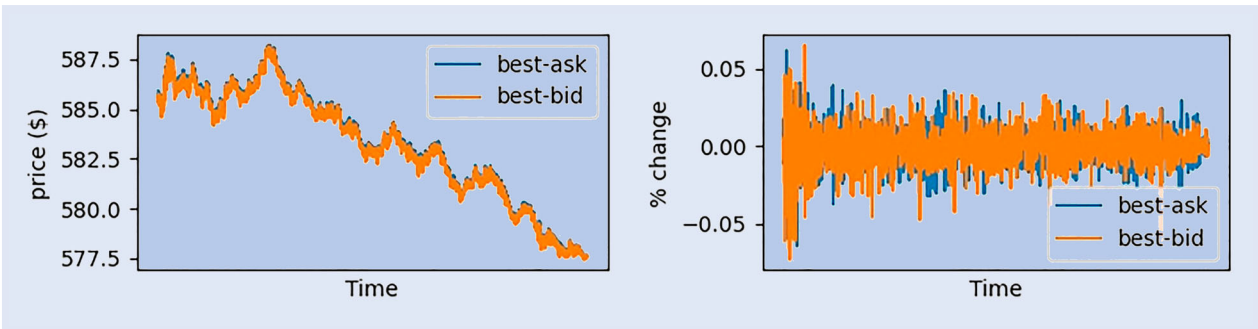

Figure 6. Price movement of a limit order book: raw time series (left) and percent-change transformed time series (right).

In our experiment, we use the Performer method (Choromanski *et al.* 2020) to enhance computational efficiency, which approximates attention with linear space and time complexity.

3.2. *Training*

3.2.1. Forecasting loss. For our evaluation, we measure the difference between predicted and actual limit orders using mean squared error (MSE) and mean absolute error (MAE) as our primary metrics. The forecasting loss specifically aims to minimize the MSE of level-5 price and volume predictions, based on scaled values within the sample dataset. The loss is computed as follows:

$$\mathcal{L}_{\text{forecast}} = \frac{1}{4IK}\left(\sum_{i=1}^{I}\sum_{j\in\{a,b\}}\sum_{k=1}^{K}\left[\left(p_{k,i}^{j\ \text{scaled}} - \hat{p}_{k,i}^{j\ \text{scaled}}\right)^2 + \left(v_{k,i}^{j\ \text{scaled}} - \hat{v}_{k,i}^{j\ \text{scaled}}\right)^2\right]\right), \tag{8}$$

where the average is computed over all $4IK$ squared errors, with I snapshots and K levels, accounting for both price and volume across bid and ask entries. MSE is chosen because of its effectiveness in addressing multivariate forecasting tasks, particularly in transformer-based models, and its ability to reduce significant discrepancies in predictions. Although metrics like MAE are recorded for reference, they are not used as loss functions.

3.2.2. Structural regularizer. To ensure that our model incorporates the multi-level LOB structures during the training process, we introduce a structural regularizer. For instance, the level-1 ask (bid) price is defined as the lowest (highest) offer price available from sellers (buyers) quoting a security, while level-2 prices represent the second-best offers, and so forth. Let p_{ki}^{a} (p_{ki}^{b}) represent the level-k ask (bid) price for the ith snapshot, where k denotes the order level. For any fixed ith LOB snapshot, the multi-level price structure is characterized by the conditions in (1)–(3).

A Spatiotemporal sequence encoding (Spacetimeformer)

	AAPL level-1-ask-price	AAPL level-1-ask-volume	AAPL level-1-bid-price	AAPL level-1-bid-volume	AAPL level-2-ask-price	AAPL level-2-ask-volume	AAPL level-2-bid-price	AAPL level-2-bid-volume	...	Timestamp
Value	$585.68	900	$585.50	18	$585.8	100	$585.81	18	...	2012-06-21 09:30:05
Spatiotemporal embedding	0	1	2	3	4	5	6	7	...	

B Compound multivariate sequence encoding (ours)

		AAPL								...	Timestamp
		Level-1				Level-2				...	
		Ask		Bid		Ask		Bid		...	
		Price	Volume	Price	Volume	Price	Volume	Price	Volume	...	
Value		$585.68	900	$585.50	18	$585.8	100	$585.81	18	...	2012-06-21 09:30:05
Compound multivariate embedding	Type (i)	0	0	1	1	0	0	1	1	...	
	Feature (j)	0	1	0	1	0	1	0	1	...	
	Level (k)	0	0	0	0	1	1	1	1	...	
	Stock (l)	0	0	0	0	0	0	0	0	...	

Figure 7. Input encoding pipeline for limit order books: spatiotemporal (a) and compound multivariate (b) embedding. The spatiotemporal embedding uses a single embedding layer to represent variable-level information, while the compound multivariate embedding employs multiple embedding layers to separately encode each repeated feature within the variable dimension.

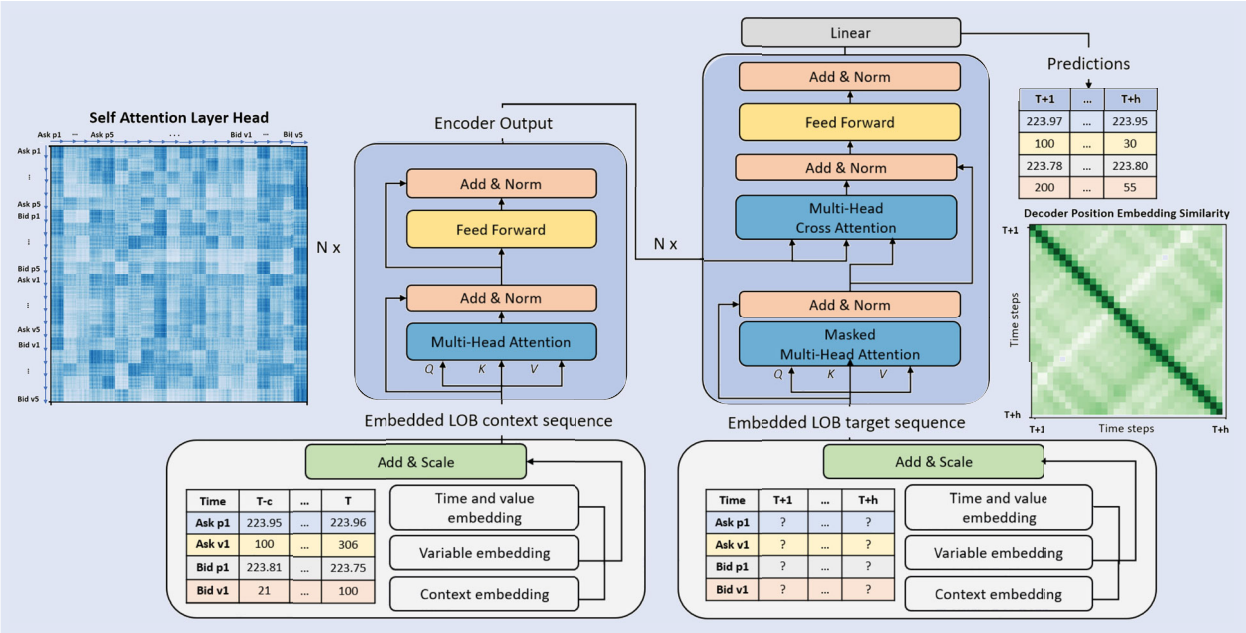

Figure 8. The spatiotemporal attention architecture is designed to forecast sequences of limit order books. It employs multiple attention modules to assign importance weights to different parts of the input, where darker shades in the attention maps indicate higher attention scores. A heatmap representation of the self-attention matrix (left, in blue) and decoder positional embedding similarity (right, in green) is included as an example snapshot to illustrate how the model attends to temporal and structural features. This approach is based on the work of Grigsby *et al.* (2022).

To nudge our network outputs adhere to these ordinal price structures, we add a regularizing term to capture this restriction at each snapshot-i:

$$\mathcal{R}_{\text{ordinal},i} = \sum_{k=1}^{K-1} \left[\text{ReLU}(\hat{p}^a_{k,i} - \hat{p}^a_{k+1,i}) + \text{ReLU}(\hat{p}^b_{k+1,i} - \hat{p}^b_{k,i})\right] + \text{ReLU}(\hat{p}^b_{1,i} - \hat{p}^a_{1,i}), \tag{9}$$

where $\hat{p}^j_{k,i}$ refers to the predicted price, and the ReLU (Rectified Linear Unit) function is defined as $\text{ReLU}(x) = \max(0, x)$, meaning that only violations (i.e. values that should be negative but are positive) contribute to the penalty. This regularization term is designed to enforce the ordinal structure of the predicted limit order book. The first two terms inside the summation ensure that ask prices increase with level, as stated in (1), and bid prices decrease with level, as stated in (3). The

Table 3. Prediction results for stock test data: Mean Squared Error (MSE) and Mean Absolute Error (MAE) are computed for each method across selected dimensions.

		Time series models		Attention-based models		
		Linear	LSTM	Temporal	Spacetime	Compound
mid-price	MSE	0.0125	0.0124	0.0124	0.0122	**0.0120**
	MAE	0.1501	0.1466	0.1520	0.1425	**0.1388**
price*	MSE	0.0026	0.0026	0.0027	**0.0025**	**0.0025**
	MAE	0.0310	0.0303	0.0332	0.0293	**0.0288**
volume**	MSE	**0.0105**	0.0118	0.0109	**0.0105**	0.0106
	MAE	0.0510	0.0541	0.0533	0.0515	**0.0504**
Forecasting loss		**0.0065**	0.0072	0.0068	**0.0065**	0.0066
Structure loss		0.2430	0.2624	0.8836	0.5774	**0.1480**
Total loss		0.0090	0.0098	0.0157	0.0123	**0.0080**

Notes: Three different embedding strategies are evaluated for attention-based models, including the compound multivariate embedding proposed in this study. Loss metrics are provided for mid-price, level-5 prices and volumes, as well as for overall forecasting and structure loss. Forecasting loss refers to MSE losses from all five levels of prices and volumes, while structure loss quantifies the violations of the ordinal structure. Total loss represents the weighted sum of the forecasting loss and structure loss. All MSE and MAE values are calculated after scaling.
* Price metrics refer to the aggregated loss for bid and ask prices across all five levels.
** Volume metrics refer to the aggregated loss for bid and ask volumes across all five levels.

Table 4. Prediction results for cryptocurrency test data.

		Attention-based models		
		Temporal	Spacetime	Compound
mid-price	MSE	0.0277	0.0272	**0.0271**
	MAE	0.1627	0.1605	**0.1600**
price*	MSE	0.0096	0.0094	**0.0093**
	MAE	0.0552	0.0544	**0.0542**
volume**	MSE	0.0012	**0.0011**	**0.0011**
	MAE	0.0136	**0.0135**	0.0136
Forecasting loss		0.0054	**0.0052**	**0.0052**
Structure loss		69.7017	29.2919	**13.9120**
Total loss		0.0123	0.0080	**0.0066**

Notes: MSE and MAE are reported across selected dimensions. Three embedding strategies are tested for attention-based models, including the proposed compound multivariate embedding. Loss metrics include mid-price, level-5 prices and volumes, forecasting loss, and structure loss. Forecasting loss captures MSE across all levels; structure loss penalizes violations of ordinal constraints. Total loss is their weighted sum. All values are computed after scaling.
* Price metrics refer to the aggregated loss for bid and ask prices across all five levels.
** Volume metrics refer to the aggregated loss for bid and ask volumes across all five levels.

final term enforces the fundamental market condition that the best bid must not exceed the best ask, as indicated in (2), preventing invalid overlaps at the top of the book.

The total loss function is defined with a regularization weight $w_o \geq 0$, which determines the strength of the regularization:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{forecast}} + w_o \cdot \sum_{i=1}^{I} \mathcal{R}_{\text{ordinal},i}, \tag{10}$$

This function penalizes predictions that break the ordinal structure of price levels while also minimizing the forecasting error, thereby ensuring that the correct multi-level sequences are maintained during training. We set $w_o = 10^{-2}$ for the stock experiments and $w_o = 10^{-4}$ for the cryptocurrency experiments to balance the relative scales of the forecasting and structure loss components. While these values are selected empirically based on the observed loss scales, w_o serves as a tunable hyperparameter that controls the penalty on structure violations and may vary depending on the characteristics of the dataset.

For training, the learning rate decay is managed with a decay factor of 0.8 and 1000 warmup steps. Additionally, the model employs seasonal decomposition and reversible normalization (Kim *et al.* 2021) techniques to enhance performance. We use three-head attention modules for training the attention-based models. We monitor the validation loss across epochs and halt the process if no improvement is observed after 10 consecutive epochs. For testing, we select the trained model weights associated with the lowest validation total loss. The training is conducted on Google Colab using a single A100 GPU.

4. Results

To assess the impact of input transformations on prediction accuracy, we first examine the results in table 2. MSE and MAE metrics are reported for different transformations applied to multi-level price data for the stock data. The combined approach, using both percent-change and min-max scaling, achieves the lowest error rates with an MSE of 0.0464 and an MAE of 0.1260, outperforming each transformation used individually. These values are presented in the raw scale to allow a consistent comparison of transformation effects.

Percent-change normalization, a standard transformation in financial time series, enforces stationarity in input sequences–an essential condition in time series forecasting. In machine learning, non-stationarity of time series can lead to unreliable predictions, as trends or shifts may skew the model's focus away from the underlying fluctuations. By transforming prices into percent changes, the model can emphasize relative

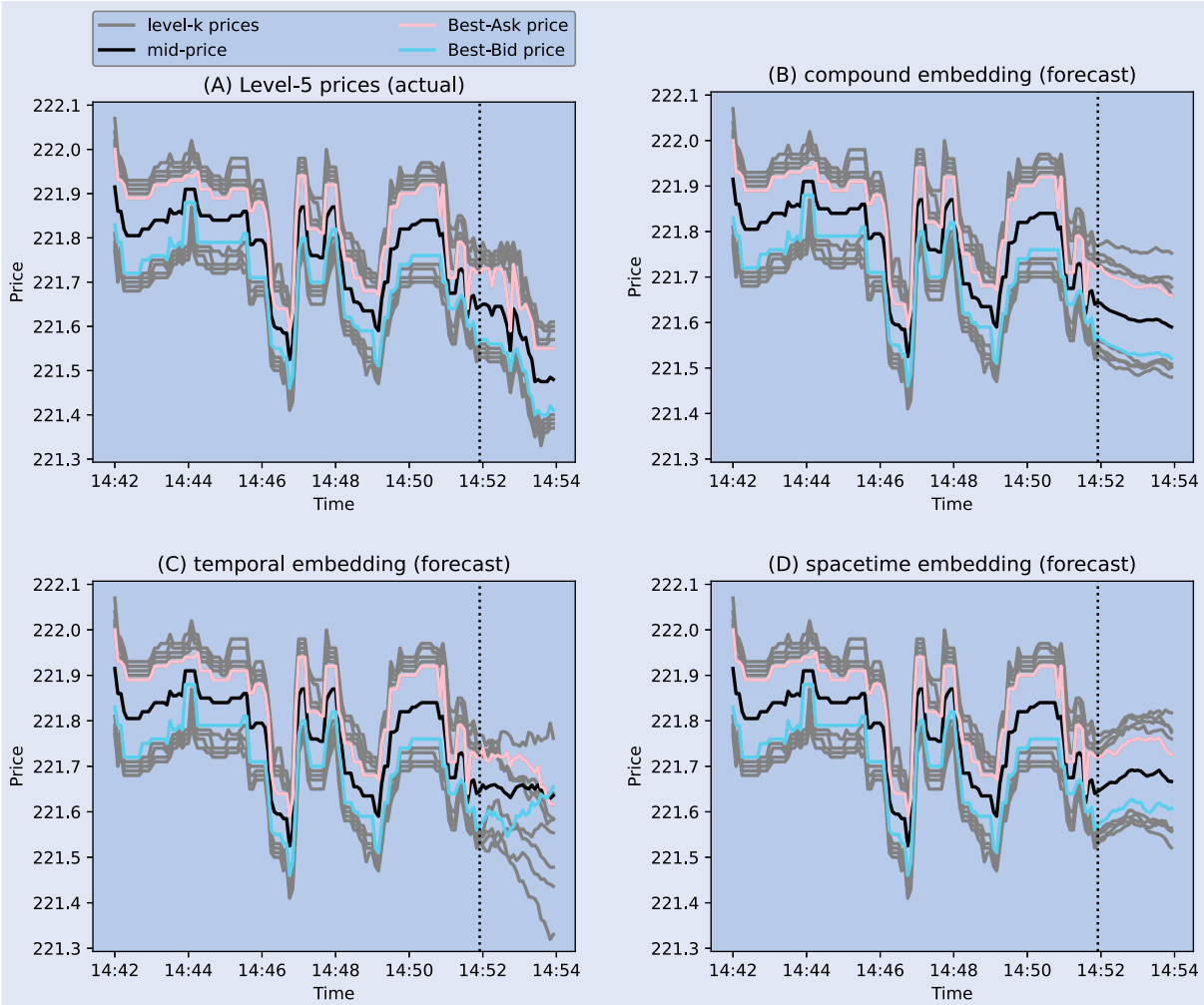

Figure 9. Price forecasting of a limit order book for AMZN stock: The graphs predict level-5 prices 2 minutes ahead, based on the preceding 10 minutes of context, with the prediction period starting on the right side of the dotted line. Panel A (top left) shows the actual price movements, while panels B, C, and D display the predicted price movements using three attention-based methods: compound multivariate embedding (B), temporal embedding (C), and spatiotemporal embedding (D). Multi-level prices, excluding the best prices, are represented in sequential grey lines.

movements around a baseline, capturing meaningful patterns without the distraction of directional trends.

Min-max scaling, on the other hand, is well-suited for multivariate data where different scales are present, as in order volume and price levels. Volume data often has higher variance than price data, and min-max scaling helps optimize learning across price levels by constraining these variations. This approach prevents extreme volume values from overwhelming the model's attention, enabling it to respond effectively to both price and volume information.

Our experimental results demonstrate that combining percent-change normalization and min-max scaling maximizes the benefits of each transformation. By stabilizing the price series for stationarity and managing volume variations within a fixed range, this combined approach achieves more accurate and consistent predictions across multi-level price data, resulting in enhanced model performance and reliability.

We now present the prediction results for test data, with a focus on various performance metrics discussed in Section 3.2.1. Our experiments compare various sequence-to-sequence prediction models of prior work, including a linear autoregressive model (*Linear*), a standard LSTM network (Schmidhuber and Hochreiter 1997) (*LSTM*), a transformer with temporal attention (*Temporal*) as introduced in Informer (Zhou *et al.* 2021), and a spacetimeformer model (*Spacetime*) that incorporates both spatial and temporal attention (Grigsby *et al.* 2022). We also evaluate the performance of our attention-based model with compound multivariate embedding (*Compound*) in comparison to the other techniques.

Table 3 compares overall prediction results of various models on level-5 stock LOB data, with the compound attention-based model (ours) showing the best overall performance. This model achieved the lowest error in all price-related metrics, including mid-price and multi-level prices, outperforming both traditional time series models and other attention-based models. Notably, the compound model excelled in maintaining ordinal structure with the lowest structure loss, leading to the lowest total loss, indicating a strong balance of accuracy and structural consistency across mid-price, price, and volume forecasting. This highlights the method's superior ability to capture structural dependencies through the compound multivariate embedding process. Although metrics calculated from mid-price,

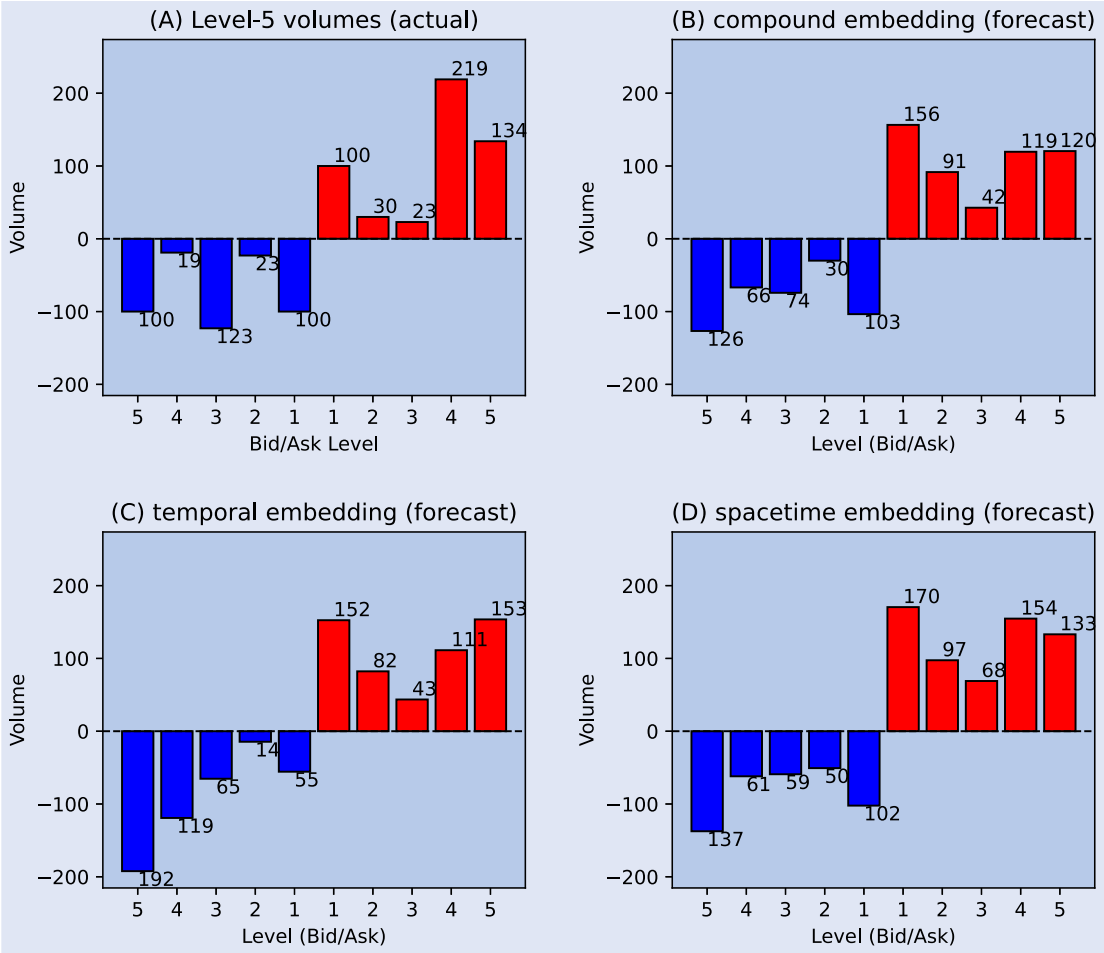

Figure 10. Volume forecasting of a limit order book snapshot for AMZN stock: The graphs forecast level-5 volumes for the first snapshot of the testing period based on the preceding 10 minutes of context. Panel A shows the actual order sizes, while panels B, C, and D display the predicted volume snapshots using three attention-based methods: compound multivariate embedding (B), temporal embedding (C), and spatiotemporal embedding (D).

multi-level prices and volumes are recorded for reference, they are not used as loss functions.

Table 4 presents the results of attention-based models on level-5 cryptocurrency LOB data. The overall trends align with those observed in the stock experiments: the proposed compound multivariate embedding (Compound) achieves the lowest structure and total loss among the three methods. While the spatiotemporal embedding (Spacetime) shows competitive forecasting performance, it fails to preserve the ordinal structure. The temporal-only embedding performs worst across all metrics. These results further demonstrate the effectiveness of the Compound embedding in modeling the structural properties of multi-level LOB data.

Figures 9 and 10 provide visual comparisons of the actual versus predicted price and volume movements for the level-5 LOB data of AMZN. We selected the first window of the test period for comparison. In figure 9, the price forecasting results demonstrate that the compound multivariate embedding method (B) closely tracks the downward trend of actual price movements (A), offering more accurate predictions than the temporal (C) and spatiotemporal (D) embedding methods. While both our approach and the spatiotemporal embedding effectively capture the overall ordinal structure of multi-level prices, the temporal embedding fails to preserve the ordinal relationships between dimensions. This aligns with the concepts of the embedding methods, as the temporal embedding processes multiple time series without incorporating variable embeddings, resulting in structural violations.

Similarly, in figure 10, the volume forecasting results indicate that the compound multivariate embedding method (B) produces a snapshot of volume distributions that align closely with the actual order sizes (A). Differences among the attention-based models are less pronounced for volume predictions than for price forecasts. Additional examples of outputs from all models, including both time series and attention-based approaches, are provided for other stocks, with results shown in figures A1, A2, A3, A4, A5, and A6.

These empirical results underscore the effectiveness of the compound multivariate embedding method in capturing the intricate dependencies within the LOB data, leading to more accurate and reliable forecasts for both price and volume.

5. Discussions

Our approach to forecasting the entire LOB is fully data-driven, allowing for predictions of both multi-level prices

and their respective order sizes at each level. This multi-level perspective expands our analytical horizon beyond mid-price forecasting, where practical applications are often limited. By capturing the granular structure of price levels, our model better reflects real market dynamics, providing a more actionable foundation for order execution.

In addition to price information, market depth–represented by the order sizes across levels–plays a significant role in influencing the LOB's rapid fluctuations. Events such as the addition, cancellation, or execution of large orders lead to priority shifts and reordering within levels, causing the LOB to dynamically adjust as orders enter or exit different price tiers. Our comprehensive, data-driven approach enables a more accurate depiction of market behavior by integrating the full scope of LOB information. Due to computational constraints, our experiments were conducted on the top five levels of the LOB. However, our method is extendable, allowing for deeper depth levels and broader time horizons.

While our model effectively captures dependencies by focusing on multi-level price and volume data alone, one limitation lies in its inability to identify the precise causes of these fluctuations–whether due to new orders, cancellations, executions, or modifications. Using an attention-based architecture, the model adapts in real time to shifts in market context, accurately tracking evolving dependencies within the LOB. However, future work could enhance this by incorporating the specific types of order events, allowing the model to differentiate between sources of LOB changes and better interpret their unique effects on LOB dynamics. Additionally, while we adopt MSE as the forecasting loss function due to its simplicity and its property of penalizing large deviations more heavily, we acknowledge that alternative metrics may offer complementary perspectives on model performance. A more comprehensive evaluation of forecasting loss functions tailored to LOB prediction remains an open direction for future research.

In practice, the ability to forecast multi-level LOB changes can significantly enhance strategies for optimal execution of large orders by simulating the LOB's response based on current data. This approach allows traders to anticipate how their trades might affect market depth, spread, and price dynamics, enabling them to adjust order placement to reduce slippage and minimize market impact. By leveraging multi-level insights, they can optimize trade size and timing to prevent adverse price moves and execute in a way that preserves liquidity.

6. Conclusions

In this study, we address the challenge of forecasting multi-level LOB data, a task that often exceeds the capabilities of conventional time-series forecasting models. Our primary contribution is the development and implementation of a compound multivariate embedding method within advanced sequence-to-sequence models. This method effectively captures the complex interdependencies among various LOB attributes, including order types, features, and levels. The empirical results demonstrate that our approach consistently outperforms existing multivariate forecasting methods, achieving the lowest forecasting errors while maintaining the ordinal structure of the LOB. These findings highlight the effectiveness of our method for financial time-series forecasting, particularly in the context of high-frequency trading. As a direction for future work, our method can be extended to address optimal execution problems, with further optimization for real-time forecasting scenarios to enhance its practical applicability in dynamic trading environments.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Supplemental data

Supplemental data for this article can be accessed online at http://dx.doi.org/10.1080/14697688.2025.2522914.

Data availability statement

The sample data that support the findings of this study are openly available in LOBSTER: Limit Order Book System - The Efficient Reconstructor at https://lobsterdata.com/. The cryptocurrency dataset is sourced from Coinbase and can be downloaded via Kaggle at https://www.kaggle.com/datasets/martinsn/high-frequency-crypto-limit-order-book-data.

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References

Abergel, F., Anane, M., Chakraborti, A., Jedidi, A. and Toke, I.M., *Limit Order Books*, 2016 (Cambridge University Press: Cambridge, UK).

Arroyo, A., Cartea, A., Moreno-Pino, F. and Zohren, S., Deep attentive survival analysis in limit order books: Estimating fill probabilities with convolutional-transformers. *Quant. Finance*, 2024, **24**, 35–57.

Bahdanau, D., Chorowski, J., Serdyuk, D., Brakel, P. and Bengio, Y., End-to-end attention-based large vocabulary speech recognition. In *Proceedings of the 2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Shanghai, China, pp. 4945–4949, 2016.

Berti, L. and Kasneci, G., TLOB: A novel transformer model with dual attention for stock price trend prediction with limit order book data. arXiv preprint arXiv:2502.15757, 2025.

Choromanski, K., Likhosherstov, V., Dohan, D., Song, X., Gane, A., Sarlos, T., Hawkins, P., Davis, J., Mohiuddin, A., Kaiser, L., Belanger, D., Colwell, L. and Weller, A., Rethinking attention with performers. arXiv preprint arXiv:2009.14794, 2020.

Chorowski, J.K., Bahdanau, D., Serdyuk, D., Cho, K. and Bengio, Y., Attention-based models for speech recognition. *Adv. Neural. Inf. Process. Syst.*, 2015, **28**, 577–585.

Cui, B., Li, Y., Chen, M. and Zhang, Z., Fine-tune BERT with sparse self-attention mechanism. In *Proceedings of the Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, Hong Kong, China, pp. 3548–3553, 2019.

Devlin, J., Chang, M.W., Lee, K. and Toutanova, K., BERT: Pre-training of deep bidirectional transformers for language understanding, 2019.

Du, S., Li, T., Yang, Y. and Horng, S.J., Multivariate time series forecasting via attention-based encoder–decoder framework. *Neurocomputing*, 2020, **388**, 269–279.

Fan, C., Zhang, Y., Pan, Y., Li, X., Zhang, C., Yuan, R., Wu, D., Wang, W., Pei, J. and Huang, H., Multi-horizon time series forecasting with temporal attention learning. In *Proceedings of the Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, Anchorage, Alaska, USA, pp. 2527–2535, 2019.

Galassi, A., Lippi, M. and Torroni, P., Attention in natural language processing. *IEEE. Trans. Neural. Netw. Learn. Syst.*, 2020, **32**, 4291–4308.

Grigsby, J., Wang, Z. and Qi, Y., Long-range transformers for dynamic spatiotemporal forecasting, 2022.

Huang, R. and Polak, T., Lobster: Limit order book reconstruction system. Available at SSRN 1977207, 2011.

Kazemi, S.M., Goel, R., Eghbali, S., Ramanan, J., Sahota, J., Thakur, S., Wu, S., Smyth, C., Poupart, P. and Brubaker, M., Time2Vec: Learning a vector representation of time, 2019.

Kercheval, A.N. and Zhang, Y., Modelling high-frequency limit order book dynamics with support vector machines. *Quant. Finance*, 2015, **15**, 1315–1329.

Kim, T., Kim, J., Tae, Y., Park, C., Choi, J.H. and Choo, J., Reversible instance normalization for accurate time-series forecasting against distribution shift. In *Proceedings of the International Conference on Learning Representations*, Vienna, Austria, 2021.

Liu, Y., Wu, H., Wang, J. and Long, M., Non-stationary transformers: Rethinking the stationarity in time series forecasting. arXiv preprint arXiv:2205.14415, 2022.

Liu, Z., Sham, C.W., Ma, L. and Fu, C., ViT-LOB: Efficient vision transformer for stockprice trend prediction using limit order books. In *Proceedings of the 2024 10th International Conference on Applied System Innovation (ICASI)*, Kyoto, Japan, pp. 436–438, 2024.

Mäkinen, Y., Kanniainen, J., Gabbouj, M. and Iosifidis, A., Forecasting jump arrivals in stock prices: New attention-based network architecture using limit order book data. *Quant. Finance*, 2019, **19**, 2033–2050.

Ntakaris, A., Magris, M., Kanniainen, J., Gabbouj, M. and Iosifidis, A., Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods. *J. Forecast.*, 2018, **37**, 852–866.

Schmidhuber, J. and Hochreiter, S., Long short-term memory. *Neural Comput.*, 1997, **9**, 1735–1780.

Shih, S.Y., Sun, F.K. and Lee, H.y., Temporal pattern attention for multivariate time series forecasting. *Mach. Learn.*, 2019, **108**, 1421–1441.

Sirignano, J.A., Deep learning for limit order books. *Quant. Finance*, 2019, **19**, 549–570.

Tsantekidis, A., Passalis, N., Tefas, A., Kanniainen, J., Gabbouj, M. and Iosifidis, A., Using deep learning for price prediction by exploiting stationary limit order book features. *Appl. Soft. Comput.*, 2020, **93**, 106401.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I., Attention is all you need. *Adv. Neural. Inf. Process. Syst.*, 2017, **30**, 5998–6008.

Wallbridge, J., Transformers for limit order books. arXiv preprint arXiv:2003.00130, 2020.

Wu, Y., Wang, S. and Fu, X., mWDN-Transformer: Utilizing cyclical patterns for short-term forecasting of limit order book in china markets. In *Proceedings of the Proceedings of the 2024 5th International Conference on Computing, Networks and Internet of Things*, Tokyo, Japan, pp. 175–181, 2024.

Yang, B., Wang, L., Wong, D.F., Shi, S. and Tu, Z., Context-aware self-attention networks for natural language processing. *Neurocomputing*, 2021, **458**, 157–169.

Zeyer, A., Irie, K., Schlüter, R. and Ney, H., Improved training of end-to-end attention models for speech recognition. arXiv preprint arXiv:1805.03294, 2018.

Zhang, Z., Zohren, S. and Roberts, S., Deeplob: Deep convolutional neural networks for limit order books. *IEEE. Trans. Signal. Process.*, 2019, **67**, 3001–3012.

Zhou, H., Zhang, S., Peng, J., Zhang, S., Li, J., Xiong, H. and Zhang, W., Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35, pp. 11106–11115, 2021.